



FATIMA JINNAH WOMEN UNIVERSITY

RAWALPINDI

Department Of Software Engineering

Cloud Computing Lab

Saira Ejaz [2023-BSE-057]

Section: 5-B

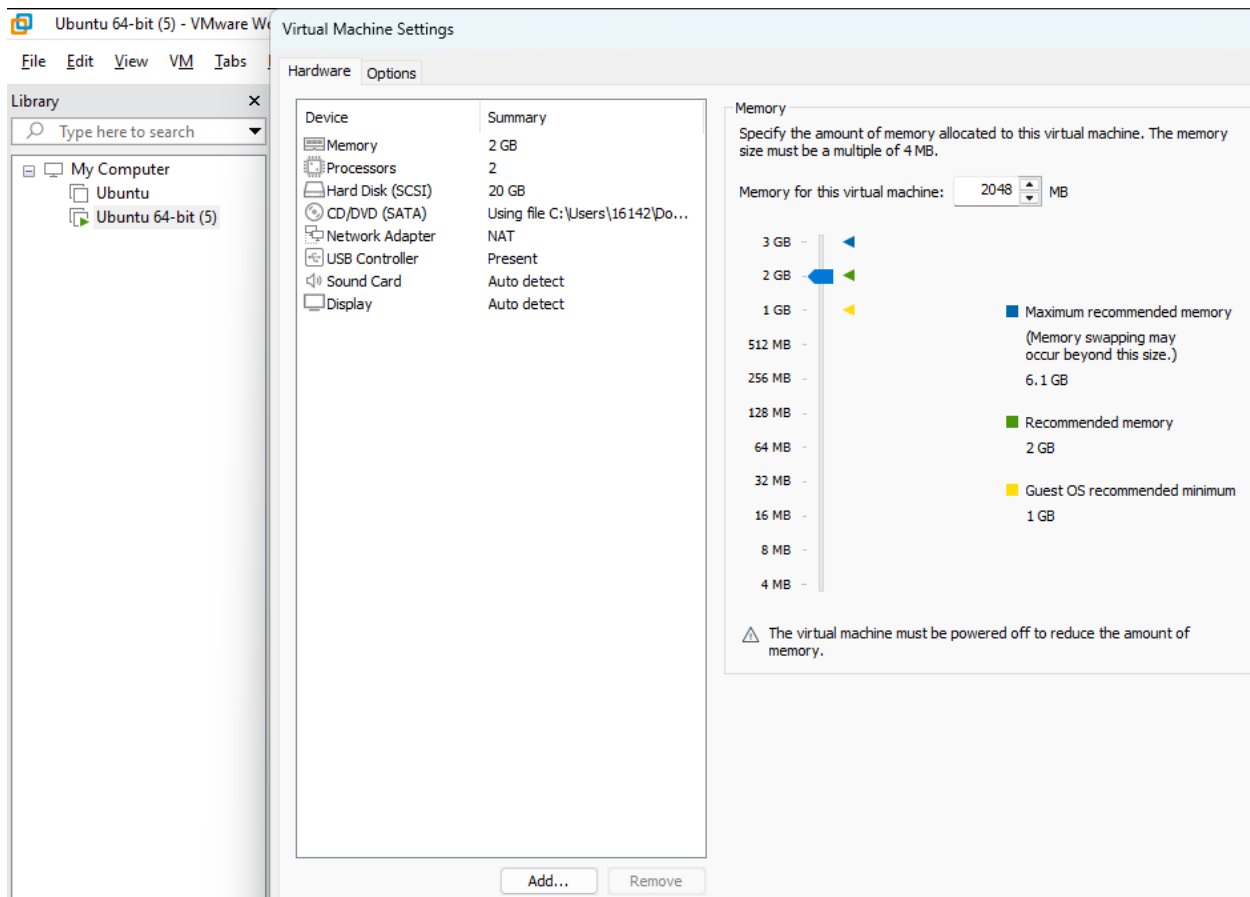
LAB NO 4

Lab Task:

Task 1 – Verify VM resources in VMware

Steps

- Open VMware Workstation and locate the Ubuntu Server VM you used in Lab 1.
- Inspect VM settings and note the following (no commands required for GUI): VM name, RAM, CPU, disk, and network adapter type



Task 2 – Start VM and log in (use your preferred host terminal method only)

Use a single preferred host-terminal method to connect to the VM. Do not switch between methods during the task.

Steps

- Start (or resume) the VM in VMware Workstation on your host.
- From your host, open your preferred terminal (for example: Windows Command Prompt, PowerShell, macOS Terminal, or Linux Terminal) and connect to the VM using SSH.

```
C:\Users\16142>ssh saira@192.168.20.128
saira@192.168.20.128's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Oct 30 11:10:00 AM UTC 2025

System load:  0.08               Processes:            218
Usage of /:   49.3% of 9.75GB    Users logged in:     1
Memory usage: 14%               IPv4 address for ens33: 192.168.20.128
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
40 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Sep 30 17:58:34 2025 from 192.168.20.1
saira@ubuntu1804:~$
```

```
saira@ubuntu1804:~$ ssh saira@192.168.20.128
saira@192.168.20.128's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Oct 30 11:12:31 AM UTC 2025

System load:  1.43           Processes:            244
Usage of /:   49.4% of 9.75GB Users logged in:       0
Memory usage: 13%          IPv4 address for ens33: 192.168.20.128
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
40 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu Oct 30 11:10:02 2025 from 192.168.20.1
saira@ubuntu1804:~$ whoami
saira
saira@ubuntu1804:~$ pwd
/home/saira
```

Task 3 – Filesystem exploration — root tree and dotfiles

Explore Linux filesystem layout and hidden files. Capture outputs using screenshots only (do not create text files). For the short explanation required in this task, write the paragraph in an editor and capture it as a screenshot (answers_md.png) — do not supply the .md file.

1. List root directory contents:

```

saira@ubuntu1804:~$ ls -la /
total 1994844
drwxr-xr-x 23 root root      4096 Sep 29 17:07 .
drwxr-xr-x 23 root root      4096 Sep 29 17:07 ..
lrwxrwxrwx  1 root root         7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x  2 root root      4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x  4 root root      4096 Sep 29 17:14 boot
dr-xr-xr-x  2 root root      4096 Aug  5 23:53 cdrom
drwxr-xr-x 20 root root     4120 Oct 30 11:12 dev
drwxr-xr-x 108 root root     4096 Oct 30 11:01 etc
drwxr-xr-x  3 root root      4096 Sep 29 17:26 home
lrwxrwxrwx  1 root root         7 Apr 22  2024 lib -> usr/lib
lrwxrwxrwx  1 root root         9 Apr 22  2024 lib64 -> usr/lib64
drwxr-xr-x  2 root root      4096 Feb 26  2024 lib.usr-is-merged
drwx-----  2 root root    16384 Sep 29 16:39 lost+found
drwxr-xr-x  2 root root      4096 Aug  5 16:54 media
drwxr-xr-x  2 root root      4096 Aug  5 16:54 mnt
drwxr-xr-x  2 root root      4096 Aug  5 16:54 opt
dr-xr-xr-x 310 root root         0 Oct 30 11:12 proc
drwx-----  4 root root      4096 Oct 30 11:07 root
drwxr-xr-x 28 root root       840 Oct 30 11:12 run
lrwxrwxrwx  1 root root         8 Apr 22  2024/sbin -> usr/sbin
drwxr-xr-x  2 root root      4096 Dec 11  2024/sbin.usr-is-merged
drwxr-xr-x  2 root root      4096 Sep 29 17:27 snap
drwxr-xr-x  2 root root      4096 Aug  5 16:54 srv
-rw-----  1 root root 2042626048 Sep 29 17:07 swap.img
dr-xr-xr-x 13 root root         0 Oct 30 11:12 sys
drwxrwxrwt 13 root root      4096 Oct 30 11:12 tmp
drwxr-xr-x 12 root root      4096 Aug  5 16:54 usr
drwxr-xr-x 13 root root      4096 Sep 29 17:25 var

```

2. Inspect these directories (run each command and screenshot the output)

ls -la /bin

```

saira@ubuntu1804:~$ ls -la /bin
lrwxrwxrwx 1 root root 7 Apr 22  2024 /bin -> usr/bin
saira@ubuntu1804:~$

```

ls -la /sbin

```

saira@ubuntu1804:~$ ls -la /sbin
lrwxrwxrwx 1 root root 8 Apr 22  2024 /sbin -> usr/sbin

```

ls -la /usr

```
saira@ubuntu1804:~$ ls -la /usr
total 96
drwxr-xr-x 12 root root 4096 Aug  5 16:54 .
drwxr-xr-x 23 root root 4096 Sep 29 17:07 ..
drwxr-xr-x  2 root root 36864 Sep 30 17:52 bin
drwxr-xr-x  2 root root 4096 Apr 22  2024 games
drwxr-xr-x 33 root root 4096 Sep 29 16:55 include
drwxr-xr-x 78 root root 4096 Sep 30 17:52 lib
drwxr-xr-x  2 root root 4096 Aug  5 17:01 lib64
drwxr-xr-x 11 root root 4096 Sep 29 16:56 libexec
drwxr-xr-x 10 root root 4096 Aug  5 16:54 local
drwxr-xr-x  2 root root 20480 Sep 30 17:52 sbin
drwxr-xr-x 124 root root 4096 Sep 30 17:52 share
drwxr-xr-x  4 root root 4096 Sep 29 16:56 src
saira@ubuntu1804:~$
```

ls -la /opt

```
saira@ubuntu1804:~$ ls -la /opt
total 8
drwxr-xr-x  2 root root 4096 Aug  5 16:54 .
drwxr-xr-x 23 root root 4096 Sep 29 17:07 ..
```

ls -la /etc

```
-rw-r--r-- 1 root root      0 Aug  5 16:54 subuid-
-rw-r--r-- 1 root root 4343 Jun 25 12:42 sudo.conf
-r--r----- 1 root root 1800 Jan 29  2024 sudoers
drwxr-xr-x  2 root root 4096 Aug  5 17:02 sudoers.d
-rw-r--r-- 1 root root 9804 Jun 25 12:42 sudo_logsrvd.conf
drwxr-xr-x  2 root root 4096 Aug  5 17:14 supercat
-rw-r--r-- 1 root root 2209 Mar 24  2024 sysctl.conf
drwxr-xr-x  2 root root 4096 Aug  5 17:02 sysctl.d
drwxr-xr-x  2 root root 4096 Aug  5 17:14 sysstat
drwxr-xr-x  6 root root 4096 Aug  5 16:49 systemd
drwxr-xr-x  2 root root 4096 Aug  5 17:00 terminfo
drwxr-xr-x  2 root root 4096 Sep 29 16:58 thermald
-rw-r--r-- 1 root root      8 Aug  5 17:02 timezone
drwxr-xr-x  2 root root 4096 Aug  5 17:14 tmpfiles.d
drwxr-xr-x  2 root root 4096 Aug  5 17:14 ubuntu-advantage
-rw-r--r-- 1 root root 1260 Jan 27  2023 ucf.conf
drwxr-xr-x  4 root root 4096 Aug  5 17:02 udev
drwxr-xr-x  2 root root 4096 Aug  5 17:14 udisks2
drwxr-xr-x  3 root root 4096 Aug  5 17:14 ufw
-rw-r--r-- 1 root root 208 Aug  5 16:54 .updated
drwxr-xr-x  3 root root 4096 Aug  5 17:02 update-manager
drwxr-xr-x  2 root root 4096 Aug  5 17:14 update-motd.d
drwxr-xr-x  2 root root 4096 Aug  5 17:14 update-notifier
drwxr-xr-x  2 root root 4096 Sep 29 16:59 UPower
-rw-r--r-- 1 root root 1523 Aug  5 17:14 usb_modeswitch.conf
drwxr-xr-x  2 root root 4096 Aug  5 17:14 usb_modeswitch.d
lrwxrwxrwx  1 root root 16 Aug  5 17:02 vconsole.conf -> default/keyboard
drwxr-xr-x  2 root root 4096 Aug  5 17:14 vim
drwxr-xr-x  4 root root 4096 Aug  5 17:14 vmware-tools
lrwxrwxrwx  1 root root 23 Feb 26  2024 vtrgb -> /etc/alternatives/vtrgb
-rw-r--r-- 1 root root 4942 Aug  5 17:14 wgetrc
drwxr-xr-x  4 root root 4096 Aug  5 17:02 X11
-rw-r--r-- 1 root root 681 Apr  8  2024 xattr.conf
drwxr-xr-x  4 root root 4096 Aug  5 17:02 xdg
drwxr-xr-x  2 root root 4096 Aug  5 17:02 xml
-rw-r--r-- 1 root root 460 Aug  5 17:14 zsh_command_not_found
saira@ubuntu1804:~$
```

ls -la /dev

```

crw-rw---- 1 root kvm      10, 124 Oct 30 11:12 udmabuf
crw----- 1 root root     10, 239 Oct 30 11:12 uhid
crw----- 1 root root     10, 223 Oct 30 11:12 uinput
crw-rw-rw- 1 root root      1,   9 Oct 30 11:12 urandom
crw----- 1 root root     10, 126 Oct 30 11:12 userfaultfd
crw----- 1 root root     10, 240 Oct 30 11:12 userio
crw-rw---- 1 root tty       7,   0 Oct 30 11:12 vcs
crw-rw---- 1 root tty       7,   1 Oct 30 11:12 vcs1
crw-rw---- 1 root tty       7,   2 Oct 30 11:12 vcs2
crw-rw---- 1 root tty       7,   3 Oct 30 11:12 vcs3
crw-rw---- 1 root tty       7,   4 Oct 30 11:12 vcs4
crw-rw---- 1 root tty       7,   5 Oct 30 11:12 vcs5
crw-rw---- 1 root tty       7,   6 Oct 30 11:12 vcs6
crw-rw---- 1 root tty       7, 128 Oct 30 11:12 vcsa
crw-rw---- 1 root tty       7, 129 Oct 30 11:12 vcsa1
crw-rw---- 1 root tty       7, 130 Oct 30 11:12 vcsa2
crw-rw---- 1 root tty       7, 131 Oct 30 11:12 vcsa3
crw-rw---- 1 root tty       7, 132 Oct 30 11:12 vcsa4
crw-rw---- 1 root tty       7, 133 Oct 30 11:12 vcsa5
crw-rw---- 1 root tty       7, 134 Oct 30 11:12 vcsa6
crw-rw---- 1 root tty       7,  64 Oct 30 11:12 vcsu
crw-rw---- 1 root tty       7,  65 Oct 30 11:12 vcsu1
crw-rw---- 1 root tty       7,  66 Oct 30 11:12 vcsu2
crw-rw---- 1 root tty       7,  67 Oct 30 11:12 vcsu3
crw-rw---- 1 root tty       7,  68 Oct 30 11:12 vcsu4
crw-rw---- 1 root tty       7,  69 Oct 30 11:12 vcsu5
crw-rw---- 1 root tty       7,  70 Oct 30 11:12 vcsu6
drwxr-xr-x 2 root root      60 Oct 30 11:12 vfiio
crw----- 1 root root     10, 127 Oct 30 11:12 vga_arbiter
crw----- 1 root root     10, 137 Oct 30 11:12 vhci
crw-rw---- 1 root kvm      10, 238 Oct 30 11:12 vhost-net
crw-rw---- 1 root kvm      10, 241 Oct 30 11:12 vhost-vsock
crw----- 1 root root     10, 122 Oct 30 11:12 vmci
crw-rw-rw- 1 root root     10, 121 Oct 30 11:12 vsock
crw-rw-rw- 1 root root      1,   5 Oct 30 11:12 zero
crw----- 1 root root     10, 249 Oct 30 11:12 zfs
saira@ubuntu1804:~$

```

ls -la /var

```

saira@ubuntu1804:~$ ls -la /var
total 56
drwxr-xr-x 13 root root    4096 Sep 29 17:25 .
drwxr-xr-x 23 root root    4096 Sep 29 17:07 ..
drwxr-xr-x  2 root root    4096 Oct 30 10:35 backups
drwxr-xr-x 16 root root    4096 Sep 30 16:39 cache
drwxrwsrwt  2 root root    4096 Aug  5 17:02 crash
drwxr-xr-x 45 root root    4096 Sep 30 16:39 lib
drwxrwsr-x  2 root staff  4096 Apr 22  2024 local
lrwxrwxrwx  1 root root      9 Aug  5 16:54 lock -> /run/lock
drwxrwxr-x 10 root syslog  4096 Oct 30 11:12 log
drwxrwsr-x  2 root mail    4096 Aug  5 16:54 mail
drwxr-xr-x  2 root root    4096 Aug  5 16:54 opt
lrwxrwxrwx  1 root root      4 Aug  5 16:54 run -> /run
drwxr-xr-x  2 root root    4096 May 21 15:46 snap
drwxr-xr-x  4 root root    4096 Aug  5 17:14 spool
drwxrwxrwt  7 root root    4096 Oct 30 11:12 tmp
-rw-r--r--  1 root root    208 Aug  5 16:54 .updated

```

ls -la /tmp

```
saira@ubuntu1804:~$ ls -la /tmp
total 52
drwxrwxrwt 13 root root 4096 Oct 30 11:12 .
drwxr-xr-x 23 root root 4096 Sep 29 17:07 ..
drwxrwxrwt 2 root root 4096 Oct 30 11:12 .font-unix
drwxrwxrwt 2 root root 4096 Oct 30 11:12 .ICE-unix
drwx----- 2 root root 4096 Oct 30 11:12 snap-private-tmp
drwx----- 3 root root 4096 Oct 30 11:12 systemd-private-e5f336b5454e4a17b290c14e2b2ae8c3-ModemManager.service-0t9FFD
drwx----- 3 root root 4096 Oct 30 11:12 systemd-private-e5f336b5454e4a17b290c14e2b2ae8c3-polkit.service-TUo0zH
drwx----- 3 root root 4096 Oct 30 11:12 systemd-private-e5f336b5454e4a17b290c14e2b2ae8c3-systemd-logind.service-0fc5oe
drwx----- 3 root root 4096 Oct 30 11:12 systemd-private-e5f336b5454e4a17b290c14e2b2ae8c3-systemd-resolved.service-FKJA3D
drwx----- 3 root root 4096 Oct 30 11:12 systemd-private-e5f336b5454e4a17b290c14e2b2ae8c3-systemd-timesyncd.service-vLytJI
drwx----- 2 root root 4096 Oct 30 11:12 vmware-root_736-2991268455
drwxrwxrwt 2 root root 4096 Oct 30 11:12 .X11-unix
drwxrwxrwt 2 root root 4096 Oct 30 11:12 .XIM-unix
```

ls -la

```
saira@ubuntu1804:~$ ls -la
total 28
drwxr-x--- 4 saira saira 4096 Sep 29 18:08 .
drwxr-xr-x 3 root root 4096 Sep 29 17:26 ..
-rw-r--r-- 1 saira saira 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 saira saira 3771 Mar 31 2024 .bashrc
drwx----- 2 saira saira 4096 Sep 29 17:33 .cache
-rw-r--r-- 1 saira saira 807 Mar 31 2024 .profile
drwx----- 2 saira saira 4096 Oct 30 11:12 .ssh
-rw-r--r-- 1 saira saira 0 Sep 29 18:08 .sudo_as_admin_successful
```

nano ~/answers.md

```
The /bin directory contain essential user commands binaries that are needed for the basic system operation such as cd and ls.
The /usr/bin directory holds most of the user commands and applications that are not required for the system to boot or repair itself.
```

```
saira@ubuntu1804:~$ cat /home/saira/answers.md
```

```
The /bin directory contain essential user commands binaries that are needed for the basic system operation such as cd and ls.
The /usr/bin directory holds most of the user commands and applications that are not required for the system to boot or repair itself.
saira@ubuntu1804:~$
saira@ubuntu1804:~$
```

Task 4 – Essential CLI tasks — navigation and file operations

Practice basic commands used daily on Linux systems. Use screenshots only for evidence.

1. Create a workspace and navigate:

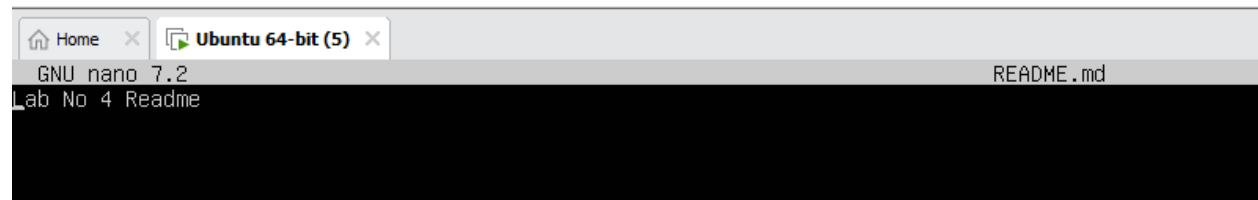
mkdir -p ~/lab4/workspace/python_project

cd ~/lab4/workspace/python_project

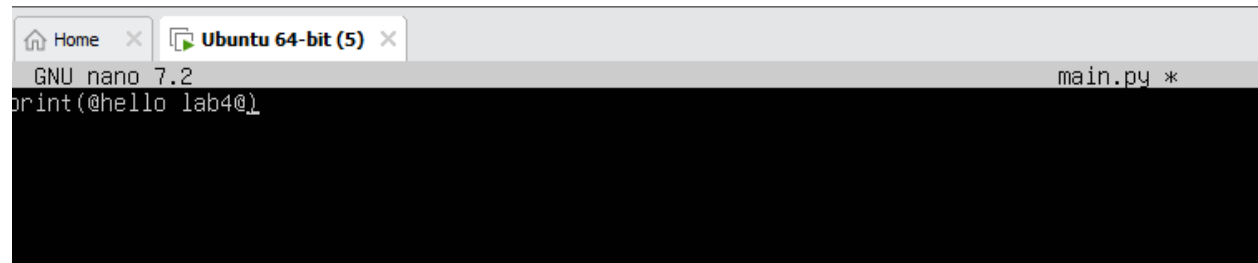
```
saira@ubuntu1804:~$ mkdir -p ~/lab4/workspace/python_project
saira@ubuntu1804:~$ cd ~/lab4/workspace/python_project
saira@ubuntu1804:~/lab4/workspace/python_project$ pwd
/home/saira/~/lab4/workspace/python_project
saira@ubuntu1804:~/lab4/workspace/python_project$ nano README.md
```

2. Create files using an editor (open each editor session and save a screenshot showing content):

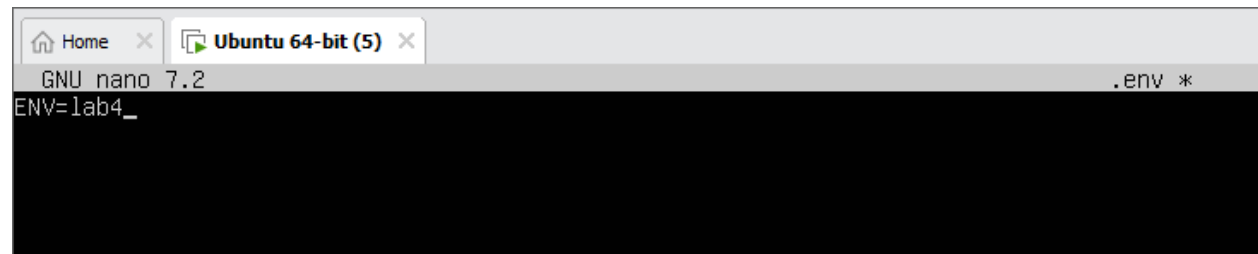
nano README.md

A screenshot of a terminal window with a nano editor session. The window title is 'Ubuntu 64-bit (5)'. The nano header shows 'GNU nano 7.2' and the filename 'README.md'. The content of the file is 'Lab No 4 Readme'.A screenshot of a terminal window showing the command 'nano main.py' being executed. The prompt is 'saira@ubuntu1804:~/lab4/workspace/python_project\$'.

nano main.py

A screenshot of a terminal window with a nano editor session. The window title is 'Ubuntu 64-bit (5)'. The nano header shows 'GNU nano 7.2' and the filename 'main.py *'. The content of the file is 'print(@hello lab4@)'.

nano .env

A screenshot of a terminal window with a nano editor session. The window title is 'Ubuntu 64-bit (5)'. The nano header shows 'GNU nano 7.2' and the filename '.env *'. The content of the file is 'ENV=lab4_'.

ls -la

3. Copy, move and remove:

cp README.md README.copy.md

```
saira@ubuntu1804:~/~/lab4/workspace/python_project$ ls -la
total 20
drwxrwxr-x 2 saira saira 4096 Oct 30 12:08 .
drwxrwxr-x 3 saira saira 4096 Oct 30 11:56 ..
-rw-rw-r-- 1 saira saira   9 Oct 30 12:08 .env
-rw-rw-r-- 1 saira saira  20 Oct 30 12:07 main.py
-rw-rw-r-- 1 saira saira  16 Oct 30 11:59 README.md
saira@ubuntu1804:~/~/lab4/workspace/python_project$ cp README.md README.copy.md
saira@ubuntu1804:~/~/lab4/workspace/python_project$ mv README.copy.md README.dev.md
saira@ubuntu1804:~/~/lab4/workspace/python_project$
```

rm README.dev.md

mkdir -p ~/lab4/workspace/java_app

cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy

ls -la ~/lab4/workspace

```
saira@ubuntu1804:~/~/lab4/workspace/python_project$ mkdir -p ~/lab4/workspace/java_app
saira@ubuntu1804:~/~/lab4/workspace/python_project$ cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy
cp: cannot stat '~/lab4/workspace/python_project': No such file or directory
saira@ubuntu1804:~/~/lab4/workspace/python_project$ cd
saira@ubuntu1804:~$ mkdir -p ~/lab4/workspace/java_app
saira@ubuntu1804:~$ cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy
saira@ubuntu1804:~$ ls -la ~/lab4/workspace
total 20
drwxrwxr-x 5 saira saira 4096 Oct 30 12:12 .
drwxrwxr-x 3 saira saira 4096 Oct 30 11:56 ..
drwxrwxr-x 2 saira saira 4096 Oct 30 12:12 java_app
drwxrwxr-x 3 saira saira 4096 Oct 30 12:12 java_app_copy
drwxrwxr-x 3 saira saira 4096 Oct 30 12:10 python_project
saira@ubuntu1804:~$
```

history

```

ls -la
16 nano /answers.md
17 nano ~/answers.md
18 nano nano /home/saira/answers.md
19 cat /home/saira/answers.md
20 mkdir -p ~/lab4/workspace/python_project
21 cd ~/lab4/workspace/python_project
22 pwd
23 nano README.md
24 cd
25 pwd
26 rm /lab4/workspace/python_project
27 rm ~/lab4/workspace/python_project
28 pwd
29 mkdir -p /lab4/workspace/python_project
30 mkdir -p ~/lab4/workspace/python_project

31 mkdir -p ~/lab4/workspace/python_project
32 cd ~/lab4/workspace/python_project
33 pwd
34 nano README.md
35 nano main.py
36 nano .env
37 ls -la
38 cp README.md README.copy.md
39 mv README.copy.md README.dev.md
40 rm README.dev.md
41 mkdir -p ~/lab4/workspace/java_app
42 cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy
43 cd
44 mkdir -p ~/lab4/workspace/java_app
45 cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy
46 ls -la ~/lab4/workspace
47 history
saira@ubuntu1804:~$

```

Task 5 – System info, resources & processes

Collect system information and observe processes. Use screenshots only.

Steps (inside VM terminal)

1. Kernel and OS:

uname -a

```

saira@ubuntu1804:~$ uname -a
Linux ubuntu1804 6.8.0-71-generic #71-Ubuntu SMP PREEMPT_DYNAMIC Tue Jul 22 16:52:38 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
saira@ubuntu1804:~$

```

2. CPU (ensure model name visible):

cat /proc/cpuinfo

```

bogomips      : 3599.99
clflush_size  : 64
cache_alignment : 64
address sizes  : 45 bits physical, 48 bits virtual
power management:

processor      : 1
vendor_id     : GenuineIntel
cpu family    : 6
model         : 142
model name    : Intel(R) Core(TM) i5-8265U CPU @ 1.60GHz
stepping      : 12
microcode     : 0xf8
cpu MHz       : 1799.999
cache size    : 6144 KB
physical id   : 2
siblings      : 1
core id       : 0
cpu cores     : 1
apicid        : 2
initial apicid : 2
fpu           : yes
fpu_exception : yes
cpuid level   : 22
wp            : yes
flags         : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtso
arch_perfmon nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt
r_aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced fsgsbase tsc_adjust bmi1 avx2 smep bmi2 inv
map clflushopt xsaveopt xsavec xgetbv1 xsaves arat md_clear flush_lld arch_capabilities
bugs          : spectre_v1 spectre_v2 spec_store_bypass swapgs itlb_multihit srbds mmio_stale_data retbleed eibrs_pbrsb gds bhi
bogomips      : 3599.99
clflush_size  : 64
cache_alignment : 64
address sizes  : 45 bits physical, 48 bits virtual
power management:

saira@ubuntu1804:~$

```

3. Memory:

free -h

```

saira@ubuntu1804:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi         447Mi        497Mi        1.2Mi        1.1Gi        1.4Gi
Swap:          1.9Gi           0B         1.9Gi
saira@ubuntu1804:~$

```

4. Disk:

df -h

```

saira@ubuntu1804:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           192M  1.3M  191M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 9.8G  5.2G  4.2G  56% /
tmpfs           960M    0  960M   0% /dev/shm
tmpfs           5.0M    0   5.0M   0% /run/lock
/dev/sda2       1.8G  100M  1.6G   7% /boot
tmpfs           192M  12K  192M   1% /run/user/1000
saira@ubuntu1804:~$
saira@ubuntu1804:~$

```

5. View OS release information:

cat /etc/os-release

```
saira@ubuntu1804:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.3 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
saira@ubuntu1804:~$
```

6. Processes (show top lines of ps output):

ps aux

```
root      816  0.0  0.6 468956 13440 ?        Ssl  11:14  0:00 /usr/libexec/udisks2/udisksd
root      823  0.0  1.1 109660 23028 ?        Ssl  11:14  0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown
root      857  0.0  0.6 392028 12544 ?        Ssl  11:14  0:00 /usr/sbin/ModemManager
syslog    858  0.0  0.3 222508 6016 ?        Ssl  11:14  0:00 /usr/sbin/rsyslogd -n -iNONE
root      871  0.0  0.1 6824 2688 ?         Ss   11:14  0:00 /usr/sbin/cron -f -P
root      875  0.0  0.0 0 0 ?         S    11:14  0:00 [irq/16-vmwgfx]
root      876  0.0  0.0 0 0 ?         I<   11:14  0:00 [kworker/R-ttm]
root      882  0.0  0.4 12020 7936 ?        Ss   11:14  0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root      910  0.0  0.2 6980 4712 tty1      Ss   11:14  0:00 /bin/login -p --
root     1370  0.0  0.0 0 0 ?         S    11:14  0:00 [psimon]
saira     1372  0.0  0.5 20276 11392 ?        Ss   11:14  0:00 /usr/lib/systemd/systemd --user
saira     1374  0.0  0.1 21152 3520 ?         S    11:14  0:00 (sd-pam)
saira     1387  0.0  0.2 8652 5504 tty1      S    11:14  0:00 -bash
saira     1412  0.1  0.4 14616 8576 tty1      S+   11:15  0:05 ssh saira@192.168.20.128
root     1413  0.0  0.5 14960 10600 ?        Ss   11:15  0:00 sshd: saira [priv]
saira     1462  0.0  0.3 15120 6956 ?         S    11:15  0:02 sshd: saira@pts/0
saira     1463  0.0  0.2 8652 5504 pts/0    Ss   11:15  0:00 -bash
root     1545  0.0  0.4 313956 8832 ?        Ssl  11:44  0:00 /usr/libexec/upowerd
root     1592  0.0  0.0 2800 1920 ?        Ss   11:46  0:00 /bin/sh /usr/lib/apt/apt.systemd.daily update
root     1597  0.0  0.0 2800 1792 ?        S    11:46  0:00 /bin/sh /usr/lib/apt/apt.systemd.daily lock_is_held update
root     2083  0.0  0.0 0 0 ?         I<   11:47  0:00 [kworker/1:1H-kblockd]
root     2084  0.0  0.0 0 0 ?         I<   11:47  0:00 [kworker/0:2H-kblockd]
root     2087  0.0  0.0 0 0 ?         I    11:47  0:00 [kworker/u257:3-events_power_efficient]
root     2147  1.2  6.6 370680 130456 ?       Sl   11:47  0:24 /usr/bin/python3 /usr/bin/unattended-upgrade --download-only
_lapt     2206  2.2  0.5 22908 10624 ?        R    11:48  0:43 /usr/lib/apt/methods/http
_lapt     2207  0.0  0.5 22904 10624 ?        S    11:48  0:00 /usr/lib/apt/methods/http
root     2230  0.0  0.0 0 0 ?         I    12:00  0:00 [kworker/0:3-rcu_par_gp]
root     2233  0.0  0.0 0 0 ?         I    12:01  0:00 [kworker/u257:1-events_unbound]
root     2247  0.0  0.0 0 0 ?         I    12:05  0:00 [kworker/u258:1-events_power_efficient]
root     2261  0.0  0.0 0 0 ?         I    12:10  0:00 [kworker/1:2-cgwb_release]
root     2271  0.0  0.0 0 0 ?         I    12:12  0:00 [kworker/u258:3-events_power_efficient]
root     2277  0.0  0.0 0 0 ?         I    12:16  0:00 [kworker/u257:0-events_power_efficient]
root     2282  0.0  0.0 0 0 ?         I    12:17  0:00 [kworker/u258:0-events_power_efficient]
root     2290  0.0  0.0 0 0 ?         I    12:20  0:00 [kworker/1:1]
root     2291  0.0  0.0 0 0 ?         I    12:20  0:00 [kworker/1:3]
saira     2293  300  0.2 10884 4480 pts/0    R+   12:20  0:00 ps aux
saira@ubuntu1804:~$
```

Task 6 – Users and account verification (no sudo group change)

Create a non-root user and verify the account exists. This task does NOT add the created user to the sudo group.

Steps (inside VM terminal)

1. Create a new user named lab4user:

```
sudo adduser lab4user
```

```
saira@ubuntu1804:~$ sudo adduser lab4user
[sudo] password for saira:
info: Adding user `lab4user' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `lab4user' (1001) ...
info: Adding new user `lab4user' (1001) with group `lab4user (1001)' ...
info: Creating home directory `/home/lab4user' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for lab4user
Enter the new value, or press ENTER for the default
  Full Name []: Saira Ejaz
  Room Number []: 2
  Work Phone []: 1234
  Home Phone []: 12345
  Other []: 122
Is the information correct? [Y/n] y
info: Adding new user `lab4user' to supplemental / extra groups `users' ...
info: Adding user `lab4user' to group `users' ...
saira@ubuntu1804:~$
```

2. Verify the user entry:

getent passwd lab4user

```
saira@ubuntu1804:~$ getent passwd lab4user
lab4user:x:1001:1001:Saira Ejaz,2,1234,12345,122:/home/lab4user:/bin/bash
saira@ubuntu1804:~$
```

3. Switch to the new user to verify login:

su - lab4user

```
saira@ubuntu1804:~$ su - lab4user
Password:
lab4user@ubuntu1804:~$
```

4. From the new user you may attempt a sudo command to show that sudo is not available for this account (expected failure), e.g.:

sudo whoami

```
lab4user@ubuntu1804:~$ sudo whoami
[sudo] password for lab4user:
lab4user is not in the sudoers file.
lab4user@ubuntu1804:~$ sudo whoami
[sudo] password for lab4user:
lab4user is not in the sudoers file.
lab4user@ubuntu1804:~$
```

5. Return to the original user:

Exit

```
lab4user@ubuntu1804:~$ exit
logout
saira@ubuntu1804:~$
```

6. (Optional) Remove the test user when finished:

sudo deluser --remove-home lab4user

```
saira@ubuntu1804:~$ sudo deluser --remove-home lab4user
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user `lab4user' ...
saira@ubuntu1804:~$
```

Bonus Task 7 – Create a small demo script using an editor and run it

This task is optional — complete it for extra practice or extra credit. It is not required for passing the core lab tasks.

Steps (inside VM)

1. Open an editor to create the script:

nano ~/lab4/workspace/run-demo.sh

Type the following lines into the editor (manually or paste), save and exit:

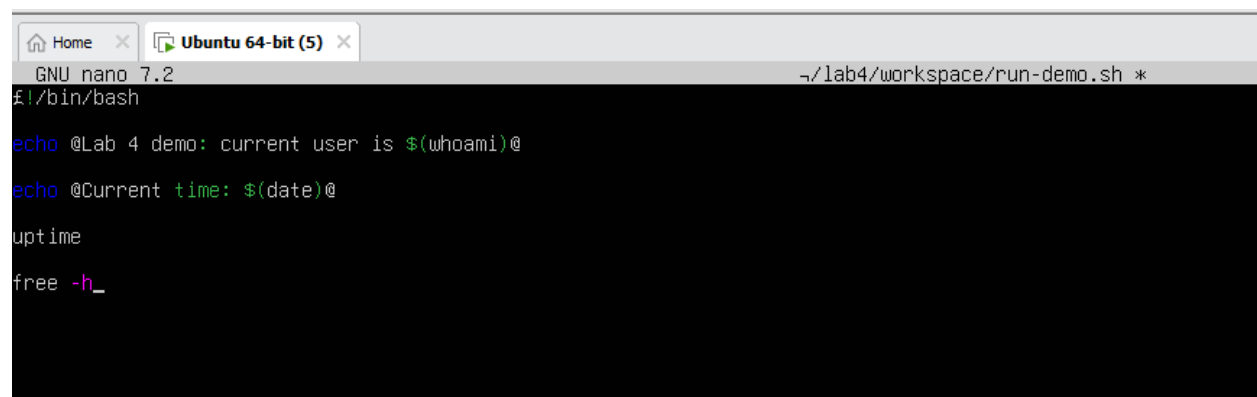
#!/bin/bash

2. **echo "Lab 4 demo: current user is \$(whoami)"**

echo "Current time: \$(date)"

uptime

free -h



```
GNU nano 7.2 ~/lab4/workspace/run-demo.sh *
#!/bin/bash

echo @Lab 4 demo: current user is $(whoami)@
echo @Current time: $(date)@
uptime
free -h_
```

```
saira@ubuntu1804:~$ cat ~/lab4/workspace/run-demo.sh
#!/bin/bash

echo @Lab 4 demo: current user is $(whoami)@

echo @Current time: $(date)@

uptime

free -h
saira@ubuntu1804:~$
```

3. Make the script executable:

chmod +x ~/lab4/workspace/run-demo.sh

```
saira@ubuntu1804:~$ chmod +x ~/lab4/workspace/run-demo.sh
saira@ubuntu1804:~$
```

4. Run the script as your regular user:

~/lab4/workspace/run-demo.sh

```
saira@ubuntu1804:~$ ~/lab4/workspace/run-demo.sh
~/lab4/workspace/run-demo.sh: line 1:#!/bin/bash: No such file or directory
@Lab 4 demo: current user is saira@
@Current time: Thu Oct 30 12:42:14 PM UTC 2025@
 12:42:14 up  1:27,  2 users,  load average: 0.00, 0.00, 0.02
            total        used        free      shared  buff/cache   available
Mem:        1.9Gi         403Mi        156Mi        1.2Mi        1.5Gi        1.5Gi
Swap:        1.9Gi           0B         1.9Gi
saira@ubuntu1804:~$
```

5. Optionally run it with sudo:

sudo ~/lab4/workspace/run-demo.sh

```
saira@ubuntu1804:~$ sudo ~/lab4/workspace/run-demo.sh
~/lab4/workspace/run-demo.sh: 1:#!/bin/bash: not found
@Lab 4 demo: current user is root@
@Current time: Thu Oct 30 12:43:00 PM UTC 2025@
 12:43:00 up  1:28,  2 users,  load average: 0.00, 0.00, 0.01
            total        used        free      shared  buff/cache   available
Mem:        1.9Gi         404Mi        155Mi        1.2Mi        1.5Gi        1.5Gi
Swap:        1.9Gi           0B         1.9Gi
saira@ubuntu1804:~$
```